William Van Eck

(717) 991-7670 | vaneck.whv@gmail.com
www.linkedin.com/in/williamvaneck | https://whvaneck.github.io

EDUCATION

The Pennsylvania State University **University Park, PA**
College of Engineering *May 2026*
Bachelor of Science in Computational Data Science

PROJECT EXPERIENCE

Poker Prediction Market **University Park, PA**
Software Engineer *January 2026 – May 2026*

- Created an interactive prediction market platform where users place probability-based bets on outcomes in simulated poker games generated through Monte Carlo bot gameplay.
- Designed dynamic contract pricing logic that updates market odds in real time based on changing probabilities of in game outcomes.
- Integrated an AI assistant that provides users with explanations, strategy, and recommendations based on the game state and scenarios available.

VisionScorer **University Park, PA**
Computer Vision Engineer *January 2025 – May 2025*

- Developed a YOLOv11 based computer vision system for basketball object detection, training models on two independently curated datasets containing 5,000 and 170 labeled images.
- Preprocessed image datasets through resizing, normalization, and augmentation to improve model consistency and reduce overfitting.
- Evaluated the impact of dataset size versus annotation quality by comparing model performance across noisy large-scale labels against smaller high quality labels using mAP performance as a metric, achieving scores of 89.9% and 75.1% respectively.
- Conducted hyperparameter tuning to optimize performance across datasets with varying sizes and label quality.

Flight Delay Prediction **University Park, PA**
Data Wrangler *November 2025 – December 2025*

- Collected and processed millions of flight records from 2015-2024 from the Bureau of Transportation Statistics, focusing on 5 airports around the State College area.
- Engineered predictive features including holiday indicators to prepare the flight dataset for machine learning models predicting delays and cancellations.
- Built and evaluated Random Forest and XGBoost classification models achieving F1 scores of .71 and .69 respectively for flight delay prediction.

WORK EXPERIENCE

Amazon **Bellefonte, PA**
Amazon Associate *February 2026 - Present*

- Processed up to 16,000 packages per day maintaining speed, accuracy, and company standards in a fulfillment environment.

UPS **Harrisburg, PA**
Driver/Loader *June 2026 – August 2026*

- Loaded and organized delivery trucks to support efficient routing, timely package distribution, and accuracy.

SKILLS

Programming: Python (Pandas, Scikit-learn, NumPy), R, SQL, Scala
Tools & Platforms: Git, Jupyter, Spark, Hadoop, HDFS, YOLO
Techniques: Machine Learning, Computer Vision, Feature Engineering, Classification, Data Preprocessing